

Invertible DeepONet: layers, sizes, parameters, training and results

Module	Layer / operation	Size in → out	Parameters
Input	design a (bounded logit of m , f_t , x , y)	8	-
	append 56 zeros	8 → 64	-
Branch T	coupling MLP: Linear + SiLU	64 → 256	16,640
10 couplings	coupling MLP: Linear + SiLU	256 → 256	65,792
	coupling MLP: Linear (log s , t)	256 → 128	32,896
	affine map $y_B = x_B e^s + t$	64 → 64	0
one coupling layer / the 10 layers			115,328 / 1,153,280
Trunk	Fourier features of f	1 → 65	0 (fixed $k\pi$)
(per frequency)	Linear + SiLU	65 → 256	16,896
	$3 \times$ [Linear + SiLU]	256 → 256	$3 \times$ 65,792
	Linear	256 → 65	16,705
trunk total			230,977
Basis	split ψ_0 (1) and $\tilde{\Psi}$ (64); QR: $\Psi^\top \Psi = FI$	301×64	0
Combine	ERP = $\Psi b + \psi_0$	$(301 \times 64)(64) \rightarrow 301$	0
Total Q64	(Q8: Q = D = 8, no padding)		1,384,257 (Q8: 938,665)
Q64 checkpoints: erp_invertible_deeponet/models/200k_2res_18modes/idon_q64.pth, idon_q64-erp.pth (same sizes); Q8: idon_q8.pth			

Sizes

design $D = 8$ (2 resonators $\times$ 4), pad 56, $Q = 64$

$F = 301$ frequencies (10-160 Hz, 0.5 Hz)

coupling: 2 hidden layers of 256, SiLU

trunk: 4 hidden layers of 256, SiLU, 32 Fourier pairs

10 couplings; masks: random halves (Q64),

even / odd alternating (Q8, $D \leq 16$)

Training (150 epochs)

data: 200k two-resonator configs, 18 plate modes,

split 160k / 20k / 20k (LHS)

Adam, lr 5×10^{-4} , cosine to 0.01 lr, clip 5, batch 128

loss = fwd + $w(e)$ Huber(inv) + 0.1 Huber(z ,0)

$w(e) = \min(1, (e + 1)/10)$; last coupling layer zero-init

Q64-ERP: fwd = MSE + 0.5 slope + 0.05 peak (from ep. 121)

Results (test; 500 targets, 16 samples)

Q8: fwd RMSE 5.98 dB, inverse ERP 9.97 dB, 24.4 cm

Q64: fwd 2.58 dB, peak err 16.7 dB, inverse 6.66 dB, 9.9 cm

Q64-ERP: fwd 3.65 dB, peak err 7.7 dB, inverse 6.80 dB, 9.9 cm

(inverse = solver ERP of the point estimate vs target;

best-of-16 oracle: 7.97 / 5.07 / 5.50 dB; Q8 early-stopped at 50)

inverse $\sim$ 0.2 ms per target (batched); no iteration